

A Template Matching Based Cough Detection Algorithm Using IMU Data From Earbuds

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Abstract—Coughing is a common symptom across different clinical conditions and has gained further relevance in the past years due to the COVID-19 pandemic. An automated cough detection for continuous health monitoring could be developed using Earbud, a wearable sensor platform with audio and inertial measurement unit (IMU) sensors. Though several previous works have investigated audio-based automated cough detection, audio-based methods can be highly power-consuming for wearable sensor applications and raise privacy concerns. In this work, we develop IMU-based cough detection using a template matching-based algorithm. IMU provides a low-power privacy-preserving solution to complement audio-based algorithms. Similarly, template matching has low computational and memory needs, suitable for on-device implementations. The proposed method uses feature transformation of IMU signal and unsupervised representative template selection to improve upon our previous work. We obtained an AUC (AUC-ROC) of 0.85 and 0.83 for cough detection in a lab-based dataset with 45 participants and a controlled free-living dataset with 15 participants, respectively. These represent an AUC improvement of 0.08 and 0.10 compared to the previous work. Additionally, we conducted an uncontrolled free-living study with 7 participants where continuous measurements over a week were obtained from each participant. Our cough detection method achieved an AUC of 0.85 in the study, indicating that the proposed IMU-based cough detection translates well to the varied challenging scenarios present in free-living conditions.

Index Terms—Cough Detection, IMU, Accelerometer, Template Matching, Machine Learning, Health Technology

I. INTRODUCTION

Coughing as a clinical symptom has gained significant public interest recently due to the COVID-19 (covid) pandemic. It is a common symptom in several clinical conditions and accounts for up to 30 million clinical visits each year, 40% of which get referred to pulmonologists [1]. Given the clinical significance of coughing, several works have investigated automated cough monitoring solutions [2], [3]. Wearable sensors are promising in this regard as they allow unobtrusive continuous monitoring.

The preliminary step for cough monitoring solutions is cough detection. Automatic cough detection alone could provide diagnostic clinical information, e.g., longitudinal information on increasing/decreasing coughing frequency. Audio-based system for automatic cough detection has been extensively investigated [3], [4]. However, an always-on audio-based solution could also raise privacy concerns. Additionally, an audio-based solution is usually power-hungry and quickly drains the wearable's battery [5]. To overcome these limitations of audio-based solutions, an inertial measurement unit (IMU) could be used to develop a stand-alone cough detection solution or a 'wake-up' cough detector to trigger the more accurate audio-based cough detection [2]. IMU sensors such as accelerometer and gyroscope can capture the typical movement signature associated with coughing to enable IMU-based cough detection.

IMU-based cough detection has been investigated in previous works. The authors in [6] developed a machine learning model-based cough detection using accelerometer features obtained from a sensor placed in the cervical/thoracic region. A similar methodology was used in [7] to detect coughs from bed-ridden participants using their smartphone's accelerometer data. In [8], an accelerometer in the epigastric region along with a neck-work strain sensor was used for cough detection. A variational autoencoder extracted the latent representation of accelerometer data which were then clustered to detect cough. Though these previous studies have shown the possibility of IMU-based cough detection, there are some limitations. For example, some of the models employed are too complex/large, e.g., with the deep learning-based autoencoders [8], which could be limiting for on-device cough detection solutions. Additionally, most of the earlier studies only considered lab-based/controlled data collection and thus could not assess performance in free-living conditions. In a previous work by our team, a template matching-based cough detection model with low computational and memory cost was proposed [5].

The study showed promising results in a lab-based and a controlled free-living dataset. However, the study employed raw accelerometer signals (x and y channel) for template matching. Different signal representations, e.g., for orientation invariance, could further improve accuracy. Additionally, the template matching-based method was not evaluated in uncontrolled free-living conditions.

In this work, we investigated template matching-based cough detection using feature representation of the IMU signal and k-medoid clustering [9] to identify representative templates. We make two main contributions. First, we propose a signal transformation of the IMU signal to improve cough detection. Our proposed approach opens avenues to investigate the combination of template matching approaches with feature-based cough detection proposed in earlier works [6], [7]. Second, we evaluated cough detection in uncontrolled free-living conditions, in addition to the evaluation under controlled conditions. Specifically, we conducted a lab-based study with 45 participants, a controlled free-living study with 15 participants, and an uncontrolled free-living study with 7 participants. Continuous measurements from up to a week were obtained from each participant in the uncontrolled free-living study. Such multi-day multi-subject free-living studies for IMU-based cough detection have not been investigated in previous works.

II. DATASET

We evaluated cough detection in three measurement settings (studies): lab-based, controlled free-living, and uncontrolled free-living. These studies were approved by the institutional review board. An ear-worn Samsung Galaxy EarBud was used in all three studies to collect audio and IMU data (50 Hz sampling frequency). The audio data were used to obtain the cough ground truths (start and end time) only [2]. We refer to [2], [5] for a detailed description of the lab-based and controlled free-living study.

A. Lab-based Cough Dataset

The lab-based cough dataset consisted of measurements from 45 participants (22 male, average age 41.4 ± 10.7 years) in 15 sessions with different user activities such as sitting, lying back, walking, stretching, etc. Eight of the sessions included voluntary coughing from the participants. On average, a session lasted for 33.4 seconds and each coughing session had 11.5 coughs with an average cough duration of 0.44 seconds.

B. Controlled Free-living Cough Dataset

The 15 participants from the lab-based study cohort with a pulmonary disease were enrolled for this study. The participants were asked to cough voluntarily in short measurement sessions in the morning and afternoon when in two different situations (sitting and walking). The measurements were obtained from multiple days (7.9 days on average), with 1.5 minutes of average recording per day.

C. Uncontrolled Free-living Cough Dataset

The uncontrolled free-living study had 7 participants from the controlled free-living study cohort. The participant wore the earbud in their free-living situation, any spontaneous coughs during the day being recorded by the earbud, with no restrictions on the participant's activities. We obtained measurements across 7 days for each participant with 19.8 hours of average recording per participant. In total, 138 hours of recording were obtained.

III. IMU-BASED COUGH DETECTION

A. Template Matching-Based Cough Detection Model

We used template matching for cough detection motivated by the previous work [5]. Template matching-based approaches have low computation and memory requirements compared to the alternative machine learning models [5]. The steps involved in our proposed cough detection model are as follows: preprocessing the IMU signal to remove artifacts, learning representative cough signals as cough templates from a training set using multi-centroid classification (MCC) [5] and k-medoid clustering, and evaluating new measurements based on dynamic time warping (DTW) distance to the learned templates. The proposed template learning and inference are represented in Figure 1

1) *Preprocessing of IMU Signals*: We resampled the IMU signal to 50 Hz, correcting for any minor drifts in sampling time and interpolating for any missing data sample. Then, the signal was low-pass filtered using a moving average filter of window size 10, followed by high-pass filtering using a Butterworth filter with a 1.5 Hz cutoff.

2) *Cough Template Generation*: We obtained cough templates for the detection model using the lab-based dataset which has a rich representation of coughing under different scenarios and from a larger participant group. First, we learned cough templates for each participant using the MCC algorithm (multi-centroid classification) [5]. Given the examples of cough and non-cough signal windows, MCC identifies the cough templates resulting in a high precision ($\geq 80\%$) at 100% recall. A high recall is specified to have a sensitive detection system. We used a window size of 400 milliseconds (nominal duration of cough as assessed from the lab-based cough dataset [5]) to extract cough and non-cough signals for MCC training. For each identified template, MCC also returns its corresponding threshold on the DTW distance metric to obtain the given overall precision and recall. All the cough templates of the participants in the training set (e.g., all except the test set participant in a leave-one-participant-out (LOPO) cross-validation) are further clustered with k-medoid clustering to obtain the final representative set of cough templates.

The methodology of this work differs from our previous work [5] in two key aspects.

- (i) First, while the previous work employed two-dimensional (x and y channel) acceleration, (acc_x, acc_y) , for template learning and evaluation, we propose to transform the signal to obtain a single channel magnitude

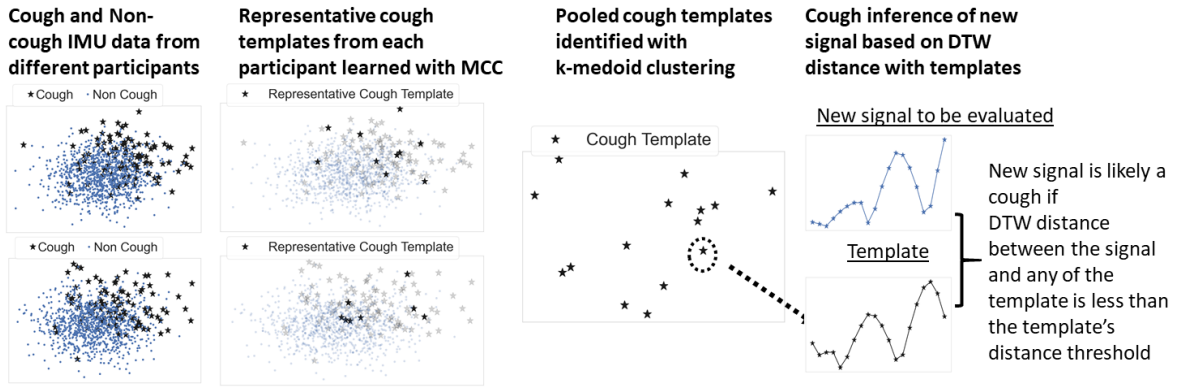


Fig. 1. Proposed IMU-based cough detection using template matching. Representative cough signals from training data earned with multi-centroid classification (MCC) are used as cough templates. A new signal window is detected as cough or not based on the DTW (Dynamic Time Warping) distance to the templates.

of jerk signal $j_{mag} = \sqrt{j_x^2 + j_y^2}$ (jerk j being the first-order derivative of accelerometer signal, $j_i = a_i - a_{i-1}$, i.e., acceleration changes). Compared to the axial accelerometer signals, a magnitude representation could provide an orientation invariant movement signal and also capture the inter-channel relations, while the signal derivative would highlight the sudden movements during coughing. X and Y channel data capture the dominant motion during cough, which we further substantiated with experiments presented in the sequel.

- (ii) Second, when identifying the cough templates from the pool of templates from training subjects, we employ k-medoid clustering that learns representative templates in an unsupervised way compared to the heuristic template ranking used in the previous work.

In this work, we show how these differences lead to significant improvement in cough detection.

3) *Inference*: Given a set of learned cough templates and their associated threshold, a new signal window is detected as cough if its DTW distance to any of the cough templates is less than the corresponding template's threshold, i.e., $distance(j_{mag}(\text{signal window}), \text{template}_k) < \text{threshold}_k$.

B. Baseline Model

We evaluated a baseline model using a one-class SVM (OC-SVM) - with an RBF kernel - trained on the statistical features of the cough templates (mean, max, median, variance, and root mean square-to-peak ratio).

C. Evaluation Metric

Similar to previous work [5], we used LOPO cross-validation to compute the ROC AUC (area under the receiver operating characteristic curve) at different detection settings (varying number of templates) for the lab-based dataset. Cough detection in the free-living datasets was evaluated using the templates learned from the lab-based dataset. Possible asynchronization between audio (modality used for cough annotations) and IMU was assessed and corrected using cross-correlation of the signals in the lab-based dataset [5]. Based on

this assessment, we set a margin of 0.5 seconds and 1 second (average and maximum drift) for asynchronization between the audio and IMU signals in the controlled and uncontrolled free-living dataset. The latter has longer recordings and might experience higher drift than the short sessions of the former. With a chosen asynchronization margin, a cough detected within the margin of annotated coughs is considered a true positive.

IV. RESULTS

A. Lab-based dataset

The AUC obtained with different cough detection methods is presented in Table I. Our proposed template matching-based method has a higher AUC than those from previous work [5], showing the improvements from the developed signal representation and template selection approach. The OC-SVM baseline model also has a lower AUC than our proposed model, indicating that the DTW distance-based template comparison is better than the template representation using chosen features in the baseline model.

TABLE I
EVALUATION OF THE PROPOSED TEMPLATE MATCHING-BASED COUGH DETECTION IN THE LAB-BASED DATASET

Model	AUC
Our proposed method - Template matching with Jerk XY magnitude and k-medoid clustering	0.85
Template matching-based cough detection from [5]	0.77
OC-SVM	0.82

We obtained 68% specificity and 90% sensitivity with our proposed method when using 50 cough templates. In comparison, the authors in [5] reported 53% specificity and 86% sensitivity. We stratified the specificity from our model in different sessions as shown in Figure 2. The specificity is low in the high-motion session such as walking, yoga stretching, and free-head movement.

We also assessed the cough detection performance obtained using 2-channel gyroscope magnitude, 2- and 3- channel

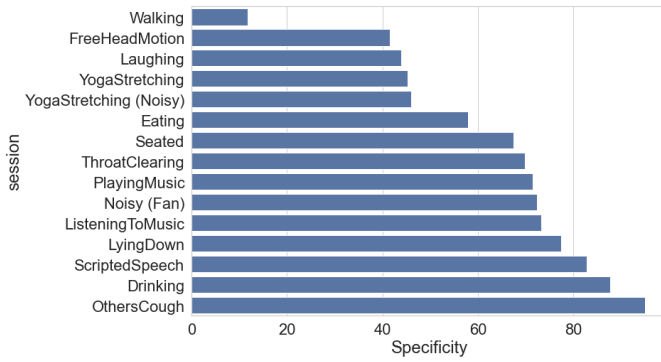


Fig. 2. Specificity of the proposed template matching-based cough detection for different sessions. The lowest specificity is obtained for the session which might produce cough-like motions in the upper body/head.

accelerometer magnitude, and 3-channel jerk magnitude as the input signal. The results obtained is shown in Table II. The accelerometer and the derived jerk signal provide better cough detection performance compared to gyroscope signals. Similarly, the z-axis movement, which captures stray motion only in the context of cough-related movements, does not improve cough detection when included in the model.

TABLE II
COUGH DETECTION PERFORMANCE IN THE LAB-BASED DATASET USING DIFFERENT INPUT SIGNALS.

Input Signal	AUC
Gyroscope X-Y Magnitude	0.76
ACC X-Y Magnitude	0.81
ACC X-Y-Z Magnitude	0.80
Jerk X-Y-Z Magnitude	0.83
Jerk X-Y Magnitude	0.85

B. Free-living Datasets

We obtained an AUC of 0.83 for cough detection in the controlled free-living dataset, compared to an AUC of 0.73 from the previous work [5]. We also evaluated a personalized cough detection model where we trained the model (obtained templates) for a participant using that participant's lab-based data. The personalized model also had an AUC of 0.83, no better than the generalized model. For the uncontrolled free-living dataset, we obtained an AUC of 0.85. The cough detection model with 50 templates had a specificity of 82% and sensitivity of 77% in this dataset.

V. DISCUSSION

For IMU-based cough detection using template matching, we found that magnitude of acceleration changes (jerk signal) as input signal improves cough detection compared to the raw 2-D acceleration used in the previous work [5] (Table I). A magnitude representation could provide orientation invariant cough signatures and acceleration changes highlight sudden movements during cough, leading to higher cough detection performance. Sessions with heavy/periodic head movement still have low specificity (Figure 2). On an earbud platform

with both IMU and audio modality, the false positives could be reduced by using audio-based cough detection [2].

In our evaluation, a personalized cough detection model did not improve upon the generalized model. This could be either due to the limited per-participant data for personalization or a reflection of the generalized nature of coughing motion across participants and needs to be investigated in future work.

We presented the first assessment of IMU-based cough detection for uncontrolled free-living conditions. The AUC obtained was comparable to those in the lab-based and controlled free-living dataset. There might be varied activities in free-living conditions but our results show that the template matching-based approach is robust to such variance. This is likely because the possible cough signatures are sufficiently represented with the templates learned from the diverse lab-based sessions. Future work could investigate the source of false positives in free-living with the inference of user's activity and improve the cough detection model accordingly.

VI. CONCLUSION

We investigated IMU-based cough detection using template matching that is amenable to resource-constrained on-device implementations, such as in earbuds. Our work shows that the magnitude of the acceleration changes (jerk) provides a better representation of motion to improve cough detection. Using this motion representation, templates identified from k-medoid clustering lead to a high AUC of cough detection across lab-based and free-living measurement conditions. IMU could thus provide a low-power privacy-preserving cough detection solution to complement audio-based cough detection.

REFERENCES

- [1] S. Sharma, M. F. Hashmi, and M. S. Alhajjaj, "Cough," in *StatPearls [Internet]*. StatPearls Publishing, 2022.
- [2] E. Nemati, S. Zhang, T. Ahmed, M. M. Rahman, J. Kuang, and A. Gao, "Coughbuddy: Multi-modal cough event detection using earbuds platform," in *2021 IEEE 17th International Conference on Wearable and Implantable Body Sensor Networks (BSN)*. IEEE, 2021, pp. 1-4.
- [3] M. Eni, V. Mordoh, and Y. Zigel, "Cough detection using a non-contact microphone: A nocturnal cough study," *Plos one*, vol. 17, no. 1, p. e0262240, 2022.
- [4] Y. Shi, H. Liu, Y. Wang, M. Cai, and W. Xu, "Theory and application of audio-based assessment of cough," *Journal of Sensors*, vol. 2018, 2018.
- [5] S. Zhang, E. Nemati, M. Dinh, N. Folkman, T. Ahmed, M. Rahman, J. Kuang, N. Alshurafa, and A. Gao, "Coughtrigger: Earbuds imu based cough detection activator using an energy-efficient sensitivity-prioritized time series classifier," in *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 1-5.
- [6] H. Mohammadi, A.-A. Samadani, C. Steele, and T. Chau, "Automatic discrimination between cough and non-cough accelerometry signal artefacts," *Biomedical Signal Processing and Control*, vol. 52, pp. 394-402, 2019.
- [7] M. Pahar, I. Miranda, A. Diacon, and T. Niesler, "Automatic non-invasive cough detection based on accelerometer and audio signals," *Journal of Signal Processing Systems*, pp. 1-15, 2022.
- [8] T. Otoshi, T. Nagano, S. Izumi, D. Hazama, N. Katsurada, M. Yamamoto, M. Tachihara, K. Kobayashi, and Y. Nishimura, "A novel automatic cough frequency monitoring system combining a triaxial accelerometer and a stretchable strain sensor," *Scientific Reports*, vol. 11, no. 1, pp. 1-9, 2021.
- [9] L. Kaufman and P. J. Rousseeuw, "Partitioning around medoids (program pam)," *Finding groups in data: an introduction to cluster analysis*, vol. 344, pp. 68-125, 1990.